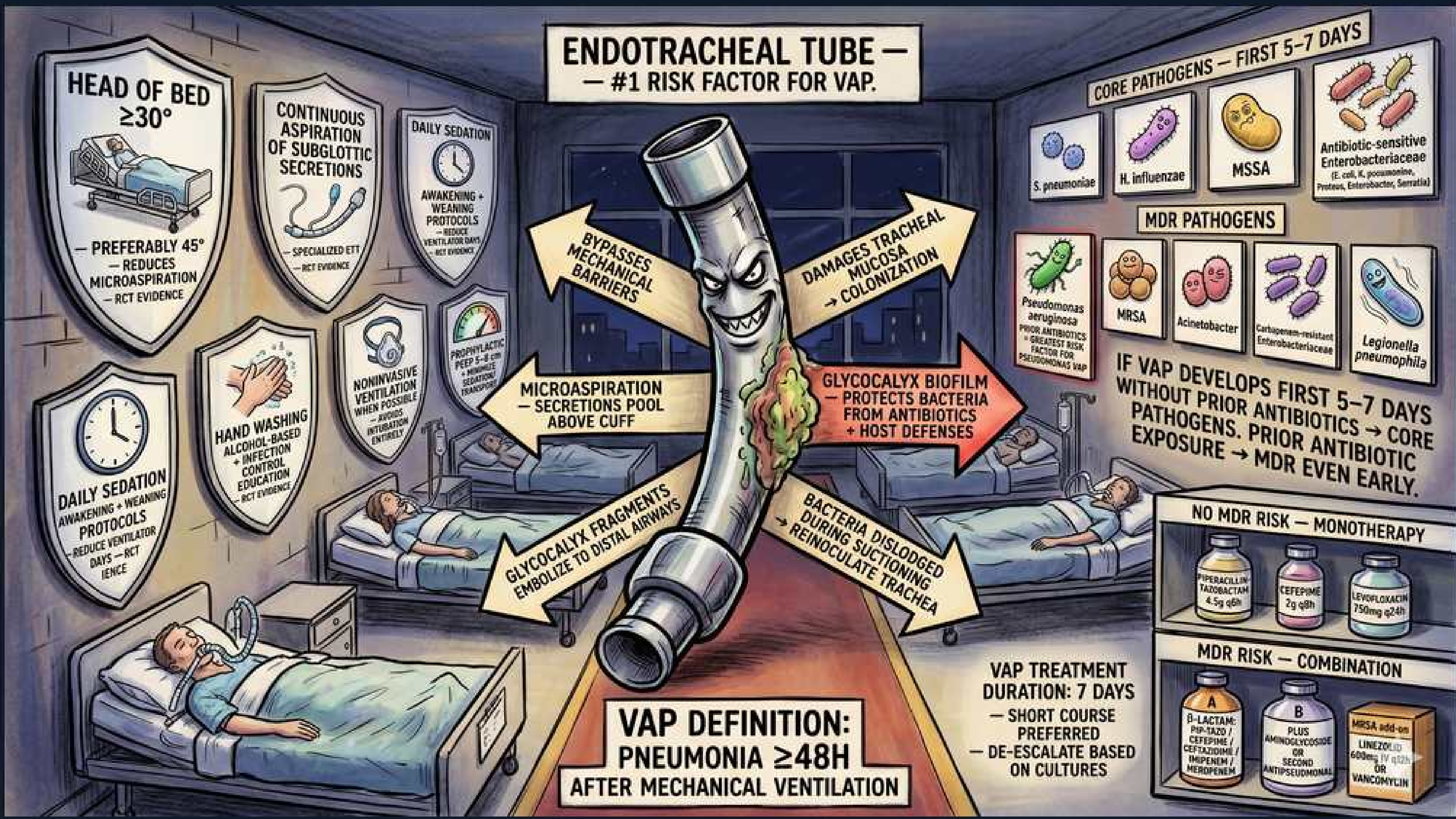
Ventilator-Associated Pneumonia: Risk, Prevention & MDR Therapy



1. ET tube = #1 VAP risk

WHAT YOU SEE
Central menacing endotracheal tube

WHAT IT ENCODES
The endotracheal tube is the single greatest risk factor for VAP: it bypasses host airway defenses, lets secretions pool above the cuff, and supports biofilm. Avoiding/limiting intubation is the best prevention.

BOARD TRAP
Non-invasive ventilation and early extubation lower VAP risk — the tube itself, not the ventilator, is the dominant driver.

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2. VAP definition >=48h

WHAT YOU SEE
Bottom banner, pneumonia >=48h after intubation

WHAT IT ENCODES
VAP is defined as pneumonia developing >=48 hours after endotracheal intubation/mechanical ventilation. Earlier infiltrates usually predate ventilation.

BOARD TRAP
Pneumonia within the first 48h of intubation is not VAP — the >=48h threshold separates VAP from community-acquired/aspiration at presentation.

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3. Head of bed >=30

WHAT YOU SEE
Top-left shield, bed elevated

WHAT IT ENCODES
Elevating the head of bed 30-45 degrees reduces gastric reflux and microaspiration, lowering VAP incidence. A core ventilator-bundle measure.

BOARD TRAP
Keeping a ventilated patient supine increases aspiration risk — semi-recumbent positioning is the evidence-based default.

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4. Subglottic suction

WHAT YOU SEE
Top shield, continuous aspiration of secretions

WHAT IT ENCODES
Continuous aspiration of subglottic secretions removes the contaminated pool above the cuff and reduces VAP. Specialized ET tubes with suction ports enable this.

BOARD TRAP
Secretions pooling above the cuff seed the lower airway — routine subglottic drainage, not just oral suctioning, addresses this reservoir.

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5. Daily sedation interruption

WHAT YOU SEE
Bottom-left shield, awakening protocol

WHAT IT ENCODES
Daily sedation interruption and spontaneous breathing trials shorten ventilation duration and thus VAP exposure. Part of the ABCDEF/ventilator bundle.

BOARD TRAP
More sedation prolongs intubation and raises VAP risk — minimizing sedation, not deepening it, is protective.

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6. Hand hygiene

WHAT YOU SEE
Shield, alcohol hand wash

WHAT IT ENCODES
Strict hand hygiene with alcohol-based rub before/after patient contact reduces cross-transmission of MDR organisms in the ICU.

BOARD TRAP
Hand hygiene remains the cheapest, highest-yield VAP-prevention intervention — never assume gloves replace it.

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7. Microaspiration

WHAT YOU SEE
Arrow, secretions pool above cuff

WHAT IT ENCODES
Microaspiration of subglottic secretions past the cuff is the principal route VAP pathogens reach the lung. Cuff pressure and subglottic suction mitigate it.

BOARD TRAP
VAP is mostly aspiration of colonized oropharyngeal secretions, not hematogenous seeding — prevention targets the airway, not bloodstream.

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8. Glycocalyx biofilm

WHAT YOU SEE
Arrow, biofilm protecting bacteria

WHAT IT ENCODES
Bacteria form a glycocalyx biofilm on the tube that shields them from antibiotics and host defenses, then dislodge to re-inoculate the trachea.

BOARD TRAP
Biofilm explains relapse despite appropriate antibiotics — source control (tube) matters because drugs penetrate biofilm poorly.

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9. Core pathogens (days 1-5)

WHAT YOU SEE
Top-right, S. pneumoniae/H. flu/MSSA

WHAT IT ENCODES
Early VAP (first 5-7 days) without prior antibiotics is caused by core, antibiotic-sensitive organisms: S. pneumoniae, H. influenzae, MSSA, and sensitive enteric GNRs. Narrow coverage often suffices.

BOARD TRAP
Early VAP in an antibiotic-naive patient does NOT need empiric MRSA/Pseudomonas coverage — that is for late or risk-factor-positive cases.

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10. MDR pathogens

WHAT YOU SEE
MRSA/Pseudomonas/Acinetobacter/Klebsiella row

WHAT IT ENCODES
Late VAP (>=5 days), prior antibiotics, or high local resistance shift the spectrum to MDR organisms: MRSA, Pseudomonas, Acinetobacter, and ESBL Klebsiella. Empiric broad coverage is warranted.

BOARD TRAP
Prior antibiotic exposure converts even early VAP to MDR risk — timing alone isn't enough; recent antibiotics escalate the regimen.

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11. Prior antibiotics shift early

WHAT YOU SEE
Middle-right note, MDR even early

WHAT IT ENCODES
If VAP develops in the first 5-7 days but the patient had prior antibiotics, treat for MDR pathogens because selection pressure has already occurred.

BOARD TRAP
Don't use a narrow early-VAP regimen in someone recently exposed to antibiotics — that exposure is the key risk modifier.

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12. No MDR risk: monotherapy

WHAT YOU SEE
Bottom-right, single agent

WHAT IT ENCODES
With no MDR risk factors, empiric monotherapy with a single antipseudomonal beta-lactam (e.g., piperacillin-tazobactam or cefepime) is acceptable for VAP.

BOARD TRAP
Double Pseudomonas coverage is not required when MDR risk is low — single appropriate agent suffices and limits toxicity.

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13. MDR risk: combination

WHAT YOU SEE
Bottom-right, two agents plus anti-MRSA

WHAT IT ENCODES
With MDR risk, use two antipseudomonal agents from different classes PLUS anti-MRSA coverage (vancomycin or linezolid), then de-escalate by culture.

BOARD TRAP
Combination therapy for MDR VAP is empiric only — once susceptibilities return, de-escalate; staying on triple coverage breeds resistance and toxicity.

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14. Treatment duration 7 days

WHAT YOU SEE
Bottom-center, duration 7 days

WHAT IT ENCODES
A short 7-day course is preferred for most VAP, with de-escalation guided by cultures and clinical response. Equivalent outcomes to longer courses with less resistance.

BOARD TRAP
Even non-fermenting GNR VAP (Pseudomonas) is now treated for ~7 days in most cases — routine 14-day courses are outdated.
